



Lab 05

COMP1323 Networks and Security

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Task i)

① Atbash Cipher:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

C R Y P T O G R A P H Y I S F U N A N D S E C U R E

X I B K G L T I Z K S B R H U F M Z M W H V X F I V (ciphertext)

② ROT 13 Cipher:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

N O P Q R S T U V W X Y Z A B C D E F G H I J K L M

C R Y P T O G R A P H Y I S F U N A N D S E C U R E

P E L C G B T E N C U L V F S H A N A Q F R P H E R (ciphertext)

③ Caesar Cipher: (shift = length of Liu Qi = 5)

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

F G H I J K L M N O P Q R S T U V W X Y Z A B C D E

C R Y P T O G R A P H Y I S F U N A N D S E C U R E

H W D U Y T L W F U M D N X K Z S F S I X J H Z W J (ciphertext)

④ Affine Cipher:

Use $E(x) = (ax + b) \bmod 26$, let $a=5$, then $b=4$

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

E J O T Y D I N S X C H M R W B G L Q V A F K P U Z

C R Y P T O G R A P H Y I S F U N A N D S E C U R E

O L U B V W I L E B N U S Q D A R E R T Q Y O A L Y (ciphertext)

ii) Polyalphabetic Cipher:

① Vigenere Cipher: key: Q I L I U

Plaintext: C R Y P T O G R A P H Y I S F U N A N D S E C U R E

Key: Q I L I U Q I L I U Q I L I U Q I L I U Q I L I U Q

ciphertext: S Z J X N E O C I J X G T A Z K V L X V I M N C L U

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

② Autokey Cipher:

key = QILIU + plaintext

QILIU C R Y P T O G R A P H Y I S F U N A N D S (keys)
 C R Y P T O G R A P H Y I S F U N A N D S E C U R E (plaintext)
 S Z J X N Q X P P I V E Z S U B L I F I M R C H U W (ciphertext)

③ Beaufort Cipher:

key: Q I L I U Q I L I U Q I L I U Q I L I U Q I L I U Q
 plaintext: C R Y P T O G R A P H Y I S F U N A N D S E C U R E
 ciphertext: O R N T B C C U I F J K D Q P W V L V R Y E J O D M

iii) Apply Transposition Techniques

① Rail-Fence Cipher

Rails = 5 (QILIU)

plaintext: C R Y P T O G R A P H Y I S F U N A N D S E C U R E

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      C           A           N           R
    R           R P       U   A       U   E
      Y       G       H       F       N       C
        P O           Y S           D E
          T           I           S
  
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ciphertext: C A N R R R P U A V E Y G H F N C P O Y S D E T I S

② Columnar Transposition Cipher:

key: Q I L I U		I I L Q U
C R Y P T	⇒	R P Y C T
O G R A P		G A R O P
H Y I S F		Y S I H F
U N A N D		N N A U D
S E C U R		E V C S R
E X X X X		X X X E X

ciphertext: R G Y N E X P A S N U X Y R I A C X C O H U S E T P F D R X

iv) Apply Matrix / Advanced Ciphers

① Playfair Cipher

Q I L U $\Rightarrow$ key

Q	I/J	L	U	A
B	C	D	E	F
G	H	K	M	N
O	P	R	S	T
V	W	X	Y	Z

CR YP TO GR AP HY IS FU NA ND SE CU RE

DP WS OP KO IT MW UP EA TF KF YM EI SD

ciphertext:

DPWSOPKOIT MWUPEATFKFYMEISD

② ADFGX or ADFGVX Cipher

key: Q I L U A

A D F G X

A	S	Y	Q	R	N
D	M	E	V	P	T
F	C	F	O	K	G
G	I	H	L	Z	U
X	W	D	X	A	B

ciphertext:

AXAGGD XDFGADFXDAXAGDAGFGAAAXDXFDFGGAGGAADGDX

DDFXXDA

V) Apply Encoding / Fractionation Ciphers

① Polybius Square Cipher

ciphertext character: Q I L U A

Q I L U A

Q	P	H	Q	G	I
I	U	M	E	A	Y
L	L	N	O	F	D
U	X	K	R	C	V
A	S	T	Z	W	B

ciphertext:

CRYPTOGRAPHY IS FUN AND SECURE

UVULIAQQAILLQVULIUQQQIIAQAAQLUIQLIULILAAQILUV
IQULIL

② Baconian Cipher

CRYPTOGRAPHY IS FUN AND SECURE

ciphertext:

A = aaaaa I/J = abaaa R = baaaa

B = aaaab K = abaab S = baaab

C = aaaba L = ababa T = baaba

D = aaabb M = ababb U/V = baabb

E = aabaa N = abbaa W = babaa

F = aabab O = abbab X = babab

G = aabba P = abbba Y = babba

aaaab baaab bbbaa abbbb baabb abbba aabba baaab aaaaa
abbbb aabbb bbbaa abaaa baaba aabab babaa abbab aaaaa
abbab aaabb baaba aabaa aaaba babaa baaab aabaa

vi) Short Explanation

① Difference between substitution and transposition ciphers:

Substitution cipher will change letters ^{and} numbers based on a rule or key. For example, it will change A to Z.

But transposition cipher will keep the letters the same but changes their physical order to hide the text.

② Advantage of using multiple encryption techniques.

v w x y z